

```
[PAXOS][LEADER=replica1] PROPOSE(1, {'op': 'touch', 'filename': 'report.txt'})
[PAXOS][replica1] ACCEPT(1, touch)
[PAXOS][replica2] ACCEPT(1, touch)
[PAXOS][replica3] ACCEPT(1, touch)
[PAXOS][replica1] LEARN(1, touch)
[PAXOS][replica2] LEARN(1, touch)
[PAXOS][replica3] LEARN(1, touch)
[PAXOS][LEADER=replica1] COMMIT(1, touch)
[PAXOS][LEADER=replica1] PROPOSE(2, {'op': 'append', 'filename': 'report.txt', 'bytes': 101, 'new_pages': 7, 'starting_page_no': 0})
[PAXOS][replica1] ACCEPT(2, append)
[PAXOS][replica2] ACCEPT(2, append)
[PAXOS][replica3] ACCEPT(2, append)
[PAXOS][replica1] LEARN(2, append)
[PAXOS][replica2] LEARN(2, append)
[PAXOS][replica3] LEARN(2, append)
[PAXOS][LEADER=replica1] COMMIT(2, append)
FILES: ['report.txt']
```

```
FULL READ:
this is a test
this project is for the cecs 327 class
we are storing pages over chord
last line here

HEAD 20 BYTES:
this is a test
this

TAIL 2 LINES:
we are storing pages over chord
last line here

Metadata for report.txt is stored at node node2 (id=190)
Page 0 for report.txt is stored at node node1 (id=154)

=== CHORD RING ===
Node node3 | id=62 | pred=227 | succ=154 | keys=[8, 37]
Node node1 | id=154 | pred=62 | succ=190 | keys=[72, 90, 111]
Node node2 | id=190 | pred=154 | succ=194 | keys=[161, 166, 172]
Node node4 | id=194 | pred=190 | succ=227 | keys=[]
Node node5 | id=227 | pred=194 | succ=62 | keys=[227]
=====
```

```
[PAXOS][LEADER=replica1] PROPOSE(3, {'op': 'delete_file', 'filename': 'report.txt', 'pages': 7})
[PAXOS][replica1] ACCEPT(3, delete_file)
[PAXOS][replica2] ACCEPT(3, delete_file)
[PAXOS][replica3] ACCEPT(3, delete_file)
[PAXOS][replica1] LEARN(3, delete_file)
[PAXOS][replica2] LEARN(3, delete_file)
[PAXOS][replica3] LEARN(3, delete_file)
[PAXOS][LEADER=replica1] COMMIT(3, delete_file)
FILES AFTER DELETE: []

=== CHORD RING ===
Node node3 | id=62 | pred=227 | succ=154 | keys=[]
Node node1 | id=154 | pred=62 | succ=190 | keys=[72]
Node node2 | id=190 | pred=154 | succ=194 | keys=[]
Node node4 | id=194 | pred=190 | succ=227 | keys=[]
Node node5 | id=227 | pred=194 | succ=62 | keys=[]
=====

[PAXOS][LEADER=replica1] PROPOSE(4, {'op': 'touch', 'filename': 'records.csv'})
[PAXOS][replica1] ACCEPT(4, touch)
[PAXOS][replica2] ACCEPT(4, touch)
[PAXOS][replica3] ACCEPT(4, touch)
[PAXOS][replica1] LEARN(4, touch)
[PAXOS][replica2] LEARN(4, touch)
[PAXOS][replica3] LEARN(4, touch)
[PAXOS][LEADER=replica1] COMMIT(4, touch)
[PAXOS][LEADER=replica1] PROPOSE(5, {'op': 'append', 'filename': 'records.csv', 'bytes': 29, 'new_pages': 2, 'starting_page_no': 0})
[PAXOS][replica1] ACCEPT(5, append)
[PAXOS][replica2] ACCEPT(5, append)
[PAXOS][replica3] ACCEPT(5, append)
[PAXOS][replica1] LEARN(5, append)
[PAXOS][replica2] LEARN(5, append)
[PAXOS][replica3] LEARN(5, append)
```

```

[PAXOS][LEADER=replica1] COMMIT(5, append)
UNSORTED INPUT:
50,z
10,a
30,x
20,b
30,alpha

[PAXOS][LEADER=replica1] PROPOSE(6, {'op': 'sort_file', 'input': 'records.csv', 'output': 'records_sorted.csv', 'records': 5})
[PAXOS][replica1] ACCEPT(6, sort_file)
[PAXOS][replica2] ACCEPT(6, sort_file)
[PAXOS][replica3] ACCEPT(6, sort_file)
[PAXOS][replica1] LEARN(6, sort_file)
[PAXOS][replica2] LEARN(6, sort_file)
[PAXOS][replica3] LEARN(6, sort_file)
[PAXOS][LEADER=replica1] COMMIT(6, sort_file)
[PAXOS][LEADER=replica1] PROPOSE(7, {'op': 'touch', 'filename': 'records_sorted.csv'})
[PAXOS][replica1] ACCEPT(7, touch)
[PAXOS][replica2] ACCEPT(7, touch)
[PAXOS][replica3] ACCEPT(7, touch)
[PAXOS][replica1] LEARN(7, touch)
[PAXOS][replica2] LEARN(7, touch)
[PAXOS][replica3] LEARN(7, touch)
[PAXOS][LEADER=replica1] COMMIT(7, touch)
[PAXOS][LEADER=replica1] PROPOSE(8, {'op': 'append', 'filename': 'records_sorted.csv', 'bytes': 29, 'new_pages': 2, 'starting_page_no': 0})
[PAXOS][replica1] ACCEPT(8, append)
[PAXOS][replica2] ACCEPT(8, append)
[PAXOS][replica3] ACCEPT(8, append)
[PAXOS][replica1] LEARN(8, append)
[PAXOS][replica2] LEARN(8, append)
[PAXOS][replica3] LEARN(8, append)
[PAXOS][LEADER=replica1] COMMIT(8, append)

```

SORTED OUTPUT:

```

10,a
20,b
30,alpha
30,x
50,z

```

STAT(records_sorted.csv):

```

{
  "filename": "records_sorted.csv",
  "size_bytes": 29,
  "num_pages": 2,
  "pages": [
    {
      "page_no": 0,
      "guid": 22
    },
    {
      "page_no": 1,
      "guid": 14
    }
  ],
  "version": 2
}

```

correctness check passed, keys: [10, 20, 30, 30, 50]